The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

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1 Introduction: The Unified 11-Part Search-and-Rescue Plot

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF). The ultimate objective of the system is the minimization of coordinate uncertainty to realize Mesarovic's Attainable Space (V'), culminating in physical vehicle extraction.

2 Axiomatic Metaphysics and the Causal Tracking Loop

The system trajectory of an asset is formalized as a higher-order relational tracking loop:

$$g(m, u, v) \equiv \exists y[R_{\text{system}}(m, u, y) \wedge E(m, u, y, v)] \tag{1}$$

where u represents environmental Navier-Stokes advection forces, m represents the external divine control input vector, y represents human or machine output behaviors, and E represents the absolute spiritual evaluation function.

2.1 Corollary 1.1: Anthropological Incompleteness and Vacuum Collapse

When the external control vector m is stripped from the network, human judgment operates as a strictly closed formal system, F. By Gödel's First Incompleteness Theorem, a closed system cannot generate its own semantic validation ($F \nvdash (v_{\text{human}} = v)$). Syntactic legalism remains blind to absolute truth, causing a critical valuation $v_k \in V'$ to become structurally unobtainable:

$$\exists v_k [V'_{v_k} \wedge \neg h(u, v_k)] \tag{2}$$

Without the active insertion of the divine control input vector, the spatiotemporal execution plane collapses into an empty relational set ($[\,]$)—a mathematical vacuum. Christ operates as the external Meta-Language that breaks this internal syntax boundary limitation.

3 The Eleven Core Structural Subsystems

3.1 The Fire-Station Dispatcher

The centralized apex root node that interfaces with the Master Functor Dispatcher (Tree #0). It accepts high-level natural language problem descriptions, checks database axioms, and triggers fleet deployment vectors.

3.2 The 50 Mobile Drone Platforms (UAVs)

A distributed fleet of autonomous double-integrator kinematic plants. Each vehicle hosts a local instance of the LISP DSL compiler and a compact neuro-symbolic engine (utilizing an LLM coding agent) to fill programmatic policy stencils (holes) on-the-fly based on immediate feedback.

3.3 The Lagrangian Weather Sensor Array

Co-moving atmospheric fluid probes mounted on each airborne platform. These probes measure local velocity vectors, feeding a Discrete Wavelet Transform (DWT) to filter high-frequency turbulence and expose scale-invariant advection trends without statistical state space estimation.

3.4 The Down-View Visual Sensor

An optical targeting payload dedicated to scanning non-convex ravine coordinates. Upon target matches, it logs the mandatory discovery passes and initializes target packet tracking.

3.5 The Communications Mesh Graph ($\mathcal{G}_C$)

A fragile topology governed by a strict 30-mile line-of-sight constraint. It maintains instantaneous mesh connectivity without store-and-forward defaults, requiring non-zero algebraic connectivity to prevent network torso fracturing ($\beta_0 > 1$).

3.6 The Computational Tasking Graph ($\mathcal{G}_{\mathbf{Comp}}$)

A network tracking local queue loads initialized near saturation limits (90%). It implements Interaction 43 cascades, meaning the battery collapse of an asset instantly escalates the computational load over surviving neighbors.

3.7 The Non-Convex Ground Obstructions

Planar 2D configuration-space keep-out zones embedded in the 100-mile grid, consisting of a Mountain Range (inducing wireless signal occlusion) and an Urban City Core (inducing bottleneck constraints).

3.8 The Battery Life & Autonomous Recharging Hubs

Non-linear power tracking variables coupled with an active infrastructure station at coordinates $[50.0, 50.0]$. If a platform's battery drops below a 30% threshold, the LISP DSL overwrites its tracking vector to execute an automatic battery swap, neutralizing Interaction 1 deletion cascades.

3.9 The Discrete Morse Filtration Engine

A combinatorial topological engine that maps a non-degenerate height filtration field $f : K \to \mathbb{R}$ over the graph complex. By evaluating face/co-face inequalities ($f(\alpha) \leq f(v)$), it pinpoints the exact coordinates of Critical Minima Traps (c_0) and Critical Saddle Edge Tunnels (c_1) like the open path $(0, 2)$.

3.10 The Self-Similar Group Controller

An algebraic scheduler governed by a Symmetric Group S_3 action. It converts spatial paths through the Morse corridors into generator words ($g_1 \cdot g_2 \cdot g_1$). Its recursive identity laws allow instantaneous inverses, supporting transactional, loss-free data rollbacks during edge-snapping events.

3.11 The Ground Rescue Vehicle

The final mechanical extraction asset. Once the store-and-forward computational buffers flush the verified visual sensor coordinates back to dispatch, the ground vehicle is targeted to the coordinates, achieving complete structural closure of the Attainable Space (V').

4 Graph Neural Networks over the Ecclesial Social Manifold

The social network layer bridges the gap between W_2 and W_1 by mapping human interaction topologies to a Graph Neural Network (GNN). Nodes represent individual human agents and edges represent relational communication channels.

4.1 Mathematical Formulation of the Message-Passing Layer

The network updates node state representations recursively across distributed overlapping simplices (the body metaphor) using a Graph Convolutional Network (GCN) update function:

$$h_i^{l+1} = \text{ReLU}\left(W^l \frac{1}{\sqrt{\deg_i}\sqrt{\deg_j}} \sum_{j \in \mathcal{N}_i} h_j^l \right) \tag{3}$$

where W^l represents the trainable layer weight tensor matrix, and $\mathcal{N}_i$ represents the local neighborhood of node i.

4.2 The Alternative Optimization Paradox

Isotropic normalization enforces Policy A (Radical Edge-Weight Equalization), stripping away ego biases. Policy B (The Matthew 18 Friction Filter) acts as a localized neighborhood pooling operation that confines high-frequency chatter to local sub-simplices, preventing non-trivial H_1 holes. To

maintain a contractible manifold ($\beta_1 = 0, \beta_0 = 1$), the global graph collapses via a mean pooling readout node to Christ, the 0-simplex anchor:

$$h_{\mathcal{G}} = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} h_i^L \tag{4}$$

This alternative optimization function balances the Inverse Centrality Reward (hub leaders expending energy to protect leaf nodes) and drives the node's existential fear parameter to zero ($\theta_i \to 0$).

4.3 Macro-System Expansion: The National Desert Wandering Orbit

When the 12 Spies generate a horizontal topological tear ($H_1 \neq 0$), the network gets trapped inside an uncollapsible horizontal echo chamber. The fault scales across the total national complex of $N = 600,000$ nodes, forcing a 40-year looping desert orbit trajectory to extract systemic variance:

$$\sigma^2(t) = \sigma_0^2 \cdot \exp(-\gamma t) \tag{5}$$

Once the generational noise is fully bled out ($\sigma^2 \to 0$), the manifold realigns to a contractible point, unblocking the Attainable Space V' for the subsequent generational layer.

5 JAX-Memo Spatial Transitions and Rational Quantitative Attribution

To model multi-agent choices under structural variation, the system integrates the JAX-accelerated state manifold over a bounded $W \times H$ lattice (Grid $= 10 \times 5$). The transition probability model is defined on a non-convex configuration space clipped by an internal boundary occlusion matrix $\mathcal{M}_{\text{wall}}$ representing physical barriers.

5.1 The Kinematic Transition Function

The forward state propagation $T(s, a, s')$ maps the continuous coordinate displacements for action vectors $a \in \mathcal{A}$:

$$T(s, a, s') = \mathbb{I}\left(s + \delta a = s' \wedge \mathcal{M}_{\text{wall}}[s'] = 0\right) + \mathbb{I}\left(s = s' \wedge \mathcal{M}_{\text{wall}}[s + \delta a] = 1\right) \tag{6}$$

where $\mathbb{I}$ is the indicator mapping function.

5.2 Cuisine Utility Attribution Vectors

A mentalizing agent computes hidden node desire parameters by marginalizing over trajectory choices using the Luce choice axiom. The probability of an output selection is proportional to the exponentiated utility matrix (wpp $= \exp(U)$):

$$P(a \mid s, \boldsymbol{\theta}) = \frac{\exp\left(\boldsymbol{\mathcal{W}}_{\text{desire}}[s] - \mathcal{C}_{\text{path}}(a)\right)}{\sum_{a' \in \mathcal{A}} \exp\left(\boldsymbol{\mathcal{W}}_{\text{desire}}[s] - \mathcal{C}_{\text{path}}(a')\right)} \tag{7}$$

By evaluating a node's path deflection around $\mathcal{M}_{\text{wall}}$ (such as discovering a hidden Lebanese truck at spot X but returning to the visible Korean truck at spot Y), the neurosymbolic compiler evaluates the commutative expectation $\mathbb{E}[\cdot]$ to uncover the favorite cuisine target attribution with zero uncertainty.